Quarter 1 Assessment Blueprint

Benchmark	MA.8.AR.2.3	MA.8.NSO.1.1	MA.8.NSO.1.2	MA.8.NSO.1.4	MA.8.NSO.1.5	MA.8.NSO.1.6	MA.8.NSO.1.3
# of Items	4	4	4	4	4	3	2
Unit of Instruction	Unit 2	Unit 2	Unit 2	Unit 3	Unit 3	Unit 3	Unit 4
2024-2025 District Percent Correct	28%	48%	31%	55%	36%	38%	29%

Total Number of Items: 25

Quarter 1
Recommended Pacing Calendar

August '25

Su	M	Tu	W	Th	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

September '25

Su	M	Tu	W	Th	F	S
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

October '25

Su	M	Tu	W	Th	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1

Quarter 1 Summary of Instructional Units

Units are linked to the SCPS Frameworks site. Bolded benchmarks are assessed on the current quarter assessment. The benchmarks that are not in bold may be assessed in the following quarter.

Unit 1 - Foundations	With 6 lessons over 9 recommended instructional days, this unit covers benchmarks MA.7.NSO.1.1, MA.7.NSO.1.2, MA.7.NSO.2.1, and MA.7.NSO.2.2. This unit is not included on the Benchmark Assessment because the benchmarks covered are below grade level. This unit builds on the foundational skills developed in Grade 7 Unit 1. In Grade 7, students rewrote positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages. Students also used the order of operations to evaluate numeric and algebraic expressions based on mathematical and real-world contexts. The expressions included all four operations, natural number exponents, and absolute value symbols. The quantities in the expressions included integers, decimals to the thousandths, positive fractions, and mixed numbers.
Unit 2 - Rational and Irrational Numbers	With 7 lessons over 10 recommended instructional days, this unit covers benchmarks **MA.8.AR.2.3, MA.8.NSO.1.1,** and **MA.8.NSO.1.2.** In this unit, students develop an understanding of irrational numbers as they continue their exploration of the real number system. In Section 1, students explore the geometric representation, algebraic notation, and verbal descriptions of perfect squares and square roots. Then they explore the geometric representations, algebraic

	notation, and verbal descriptions of perfect cubes and cube roots. Students solve equations in the form $x^2=p$ and $x^3=q$, where p is a whole number and q is an integer. They also learn to find the values of square roots and cube roots. In Section 2, students extend their knowledge of rational numbers to define and explore irrational numbers. Students use their knowledge of finding the square and cube roots of perfect squares and cubes to approximate non-perfect square and cube roots. They finish the unit by plotting, comparing, and ordering rational numbers and the approximated values of irrational numbers on a number line.
Unit 3 - **Scientific Notation**	With 8 lessons over 11 recommended instructional days, this unit covers benchmarks **MA.8.NSO.1.3, MA.8.NSO.1.4, MA.8.NSO.1.5,** and **MA.8.NSO.1.6**. In Grade 7, students developed an understanding of how to apply the laws of exponents to evaluate and simplify numerical expressions. Students also developed the knowledge and skills to perform operations with rational numbers in 7th grade. Students utilize these skills in Unit 3, as they discover how to perform operations on numbers expressed in scientific notation and express their answers in scientific notation. This work prepares students for simplifying and evaluating algebraic expressions with integer exponents, in Unit 4, and with rational number exponents in Algebra I.
Unit 4 - **Equivalent Numerical Expressions**	With 11 lessons over 16 recommended instructional days, this unit covers benchmarks **MA.8.NSO.1.3** and MA.8.AR.1.1. Assessment items aligned to **MA.8.NSO.1.3** will only be assessed as it aligns to lessons in Unit 3 and Unit 4, Lessons 1-3. In Grade 7, students developed an understanding of how to apply the Laws of Exponents to evaluate and rewrite numerical expressions. Students also developed the knowledge and skills to perform operations with rational numbers in 7th grade. Students utilized these skills in Unit 3 as they discovered how to express numbers in scientific notation and perform operations on numbers expressed in scientific notation. In this unit, students apply their prior knowledge and skills to evaluate and rewrite numeric and algebraic expressions with integer exponents. This work prepares students for simplifying and evaluating expressions with rational-number exponents in Algebra I.